ME 315 Final Project Report

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ME 315 Fall Quarter 2024

Contents

1 Executive Summary

This report explains the design for a 2 stage reducer gearbox with 60 kW of power, as shown in the diagram above. The purpose of this 2 stage reducer is to efficiently transmit power from input shaft 1 to output shaft 2, which receives power via input and output gears on the 2 shafts. The gear ratios and gearbox was designed to minimize the size of the gearbox, thereby reducing the materials and cost of the project. The system receives an input of 2700 RPM and steps it down to an output of 403.2 RPM, within the 400 $\pm$4 rpm specified, assuming 100% overall efficiency. Our design accounted for fatigue, bending, torsional, and shear stress, providing us with an overall factor of safety of 1.1, given the proposed workload/conditions of functioning at minimum 5 years, operating 52 weeks a year, 8 hours a day at 99% reliability. To solve this problem, our primary method utilizes a Python program (Linked here) that calculates the factor of safety of the shafts, bearings, and gears based on our gear design and shaft design.

2 Gear Train Design Parameters

Parameter	Gear No.			
	1	2	3	4
Module (mm)	6	6	8	8
Pressure Angle (deg)	20	20	20	20
Teeth No.	17	43	17	45
Face Width	72	72	96	96
Pitch Circle Diameter (mm)	102	258	136	360
Total Ratio	-	2.53	-	6.67
Speed (RPM)	2700	1067.44	1067.44	403.2
Power (kW)	60	60	60	60
Torque (Nm)	212.21	536.64	536.64	1420.52
Material	AISI 1030 Steel	AISI 1030 Steel	AISI 1030 Steel	AISI 1030 Steel
Young's Modulus, E (GPa)	200	200	200	200
Poisson Ratio	0.29	0.29	0.29	0.29
Hardness (HB)	401	302	401	302
FoS Contact Stress	4.95	3.72	5.25	3.95
FoS Bending Stress	1.44	1.10	1.55	1.19

3 Shaft Design & Parameters

Parameter	Shaft No.		
	1	2	3
Material	AISI 415 S41500	AISI 415 S41500	AISI 415 S41500
Ultimate Tensile Strength (MPa)	900	900	900
Yield Strength (MPa)	700	700	700
Diameter (mm)	40	48	50
Length (mm)	344	248	344
Speed (rpm)	2700	1067.44	403.2
Maximum Torque (Nm)	212.21	536.75	1420.82
Maximum Shear (N)	3232	4583	6704
Maximum Moment (Bending) (Nm)	213	323	473
Position of Critical Cross-Section (mm) (From Left)	153	165	261
Goodman FoS	1.45	1.43	1.41
Yield FoS	14.31	12.27	14.34

4 Bearing Selection & Parameters

Parameter	Bearing Position					
	A	B	C	D	E	F
Bearing No.	SKF 6407	SKF 6407	SKF 6407	SKF 6407	SKF 6407	SKF 6407
Working C (N)	37452	15923	5967	44773	7799	47350
C (N)	55300	55300	55300	55300	55300	55300
C0 (N)	31000	31000	31000	31000	31000	31000
ID/Nominal (mm)	35	35	35	35	35	35
OD (mm)	100	100	100	100	100	100
Width (mm)	25	25	25	25	25	25
Radial Load (N)	2814	1196	611	4583	1104	6704
Life (Hrs)	14560	14560	14560	14560	14560	14560
L10 (Million Cycles)	2358.72	2358.72	932.51	932.51	352.24	352.24
FoS	1.09	2.57	9.27	1.24	7.09	1.17

5 Final Results

Overall Factor of Safety (Lowest Component)	1.1 (Gear 2 Bending Stress)
Bearing Maintenance	No maintenance required, fullfils 5 year life expectancy
General Maintenance	No maintenance required, fullfils 5 year life expectancy

5.1 Shaft 1 Assembly

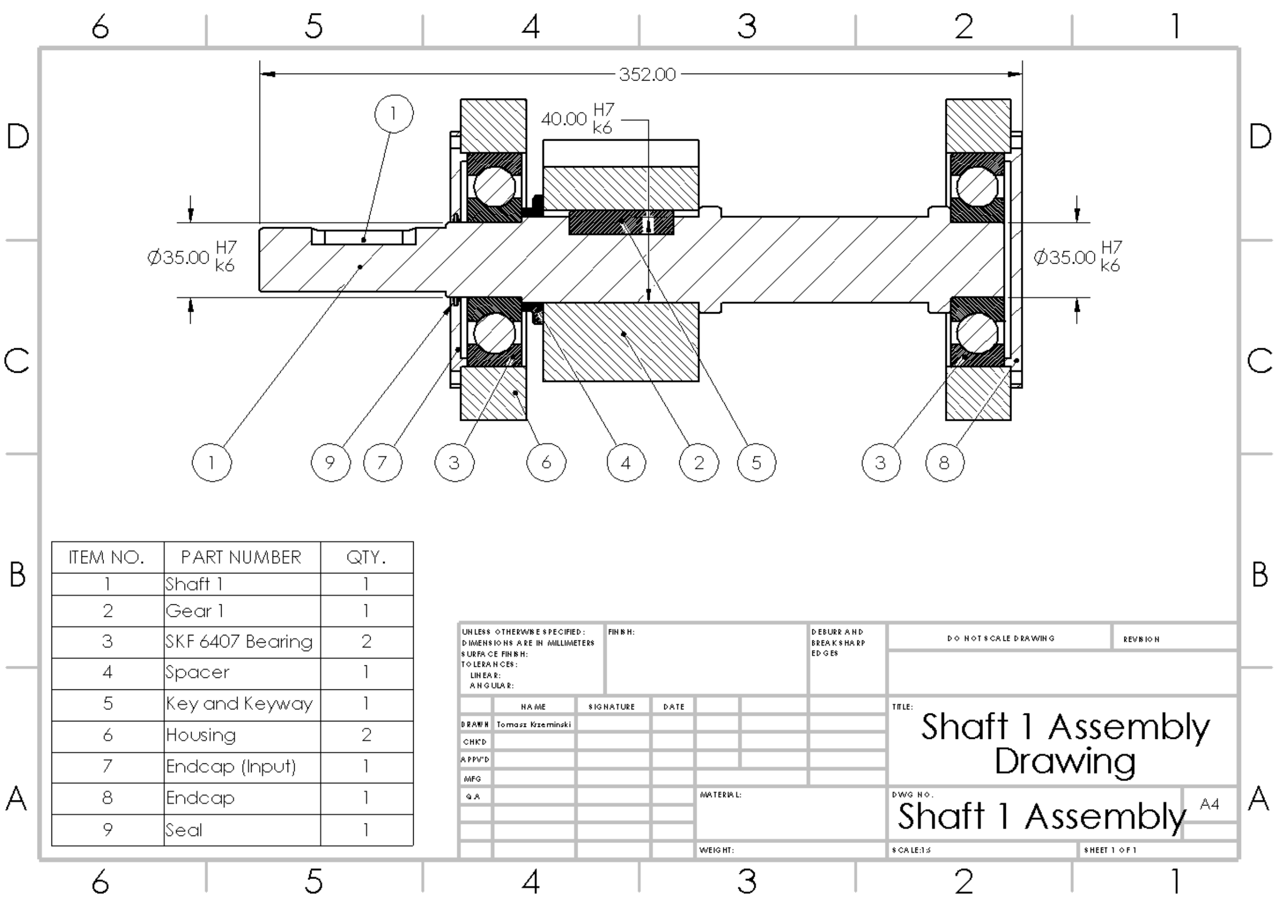

Figure 1: Shaft 1 Assembly Drawing

5.2 Shaft 2 Assembly

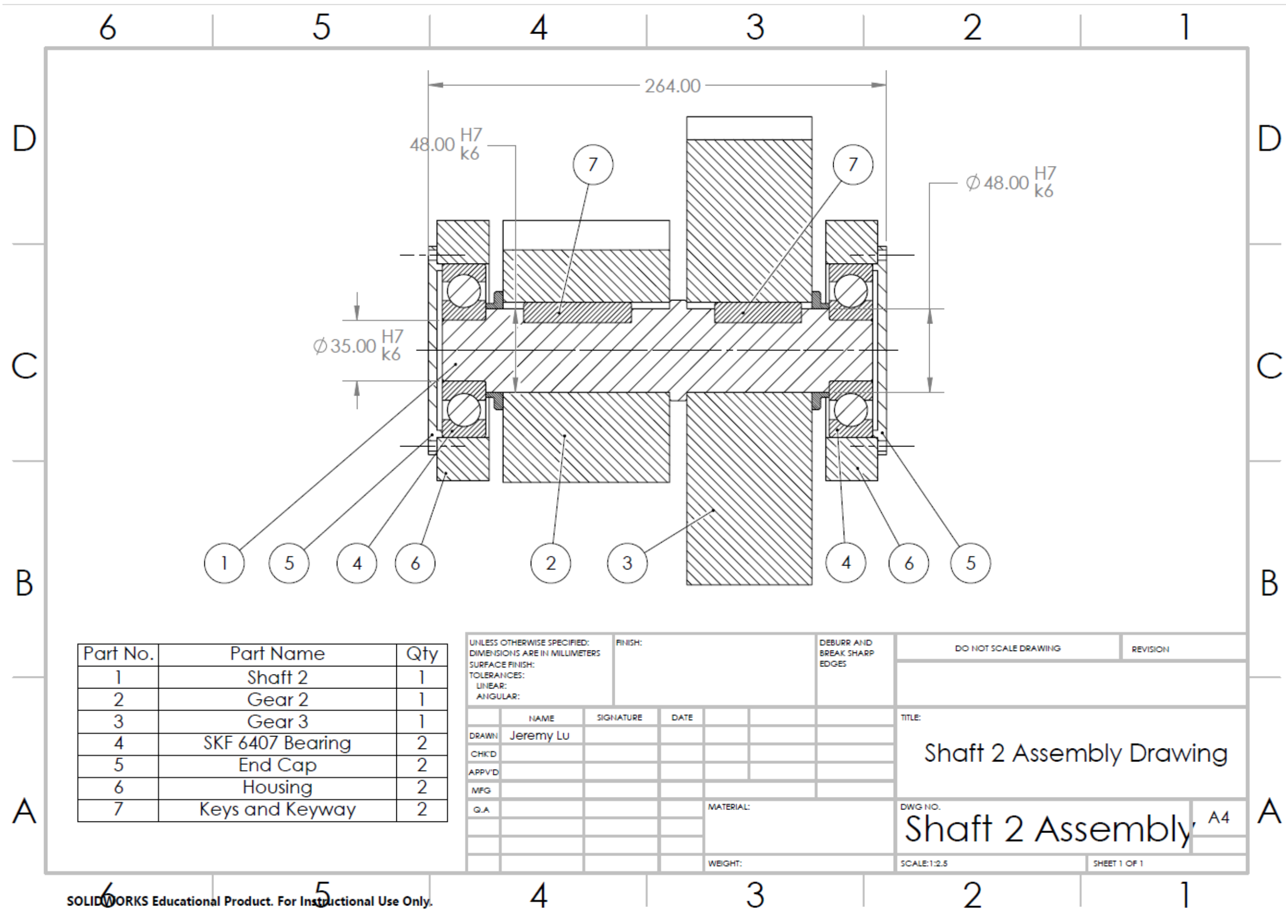

Figure 2: Shaft 2 Assembly Drawing

5.3 Shaft 3 Assembly

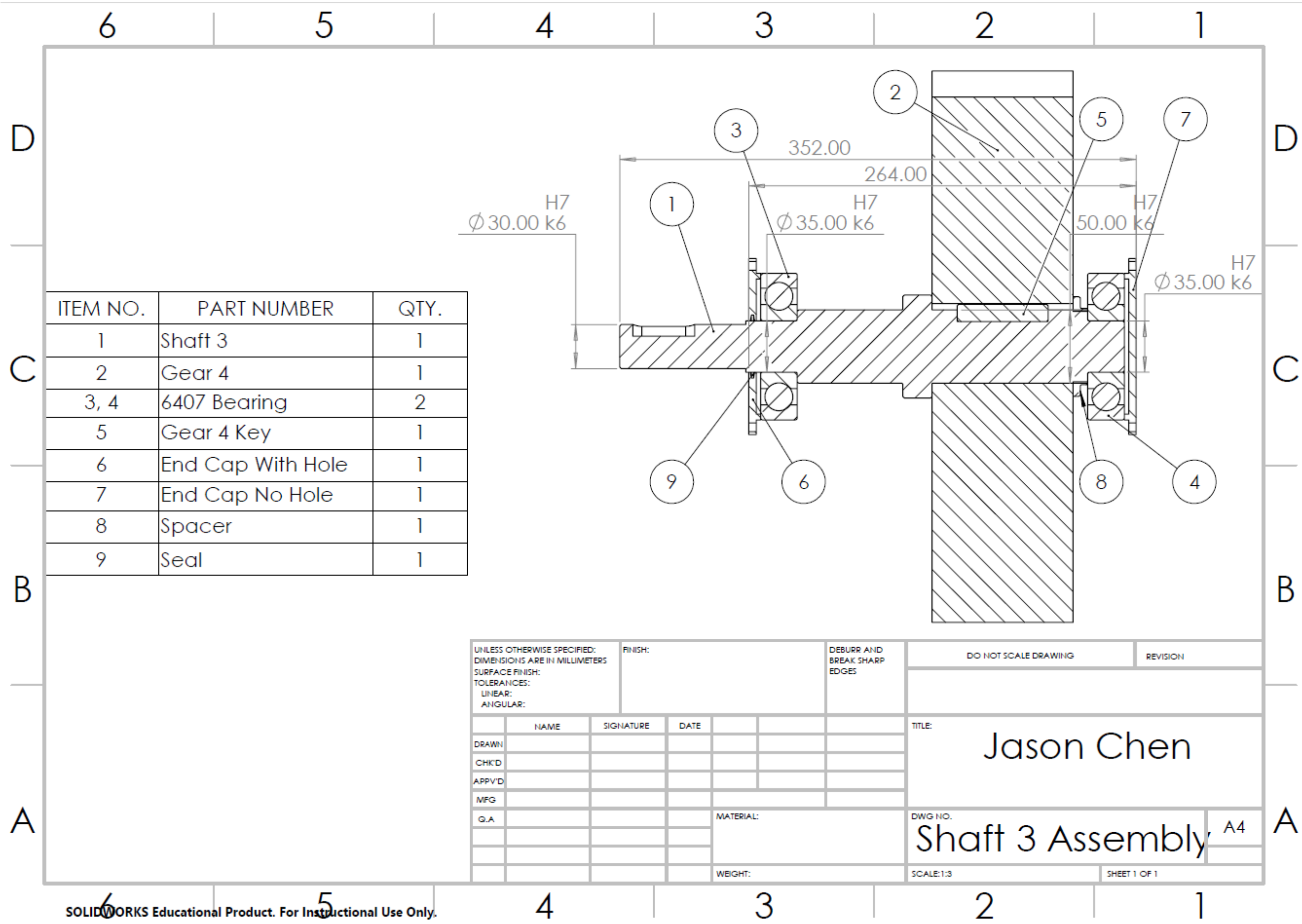

Figure 3: Shaft 3 Assembly Drawing

6 Shaft Drawings

6.1 Shaft 1

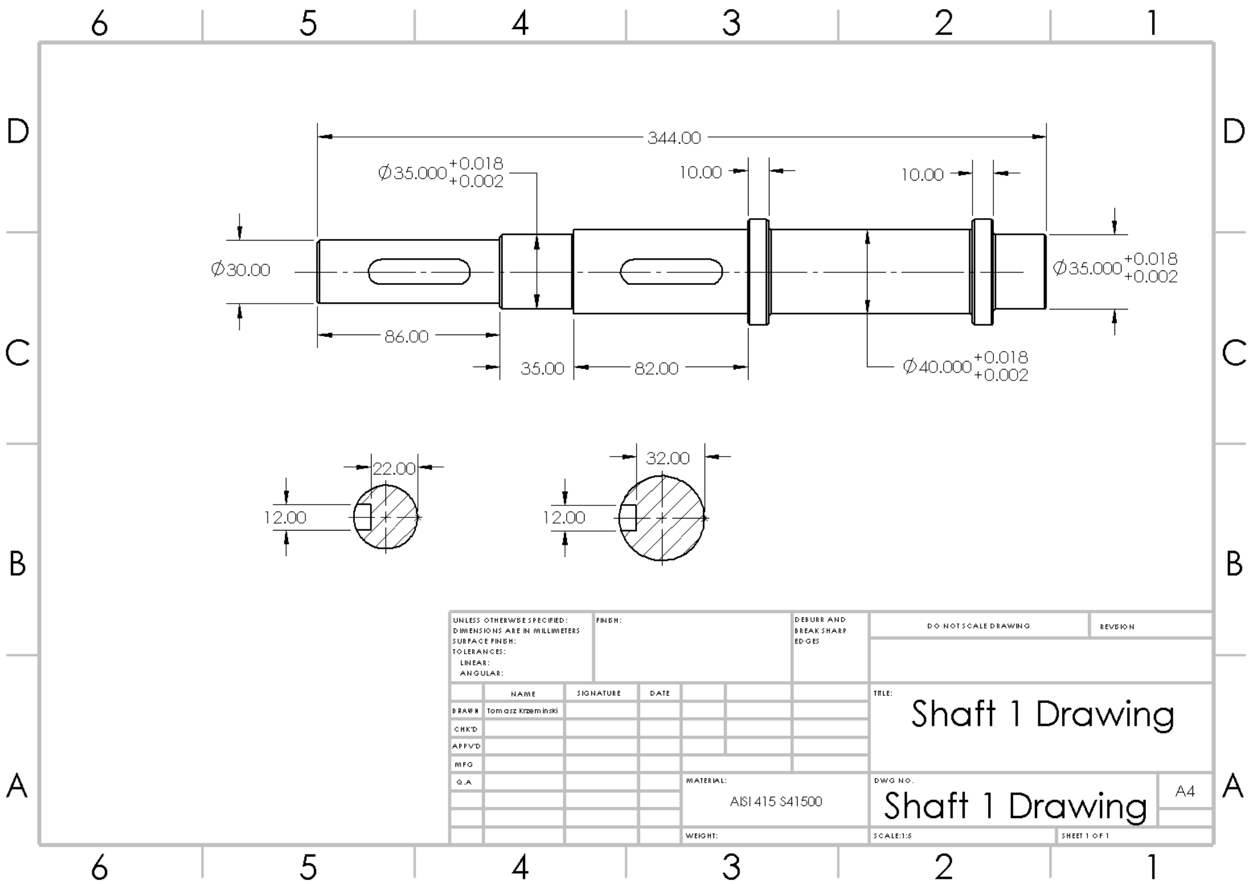

Figure 4: Shaft 1 Drawing

6.2 Shaft 2

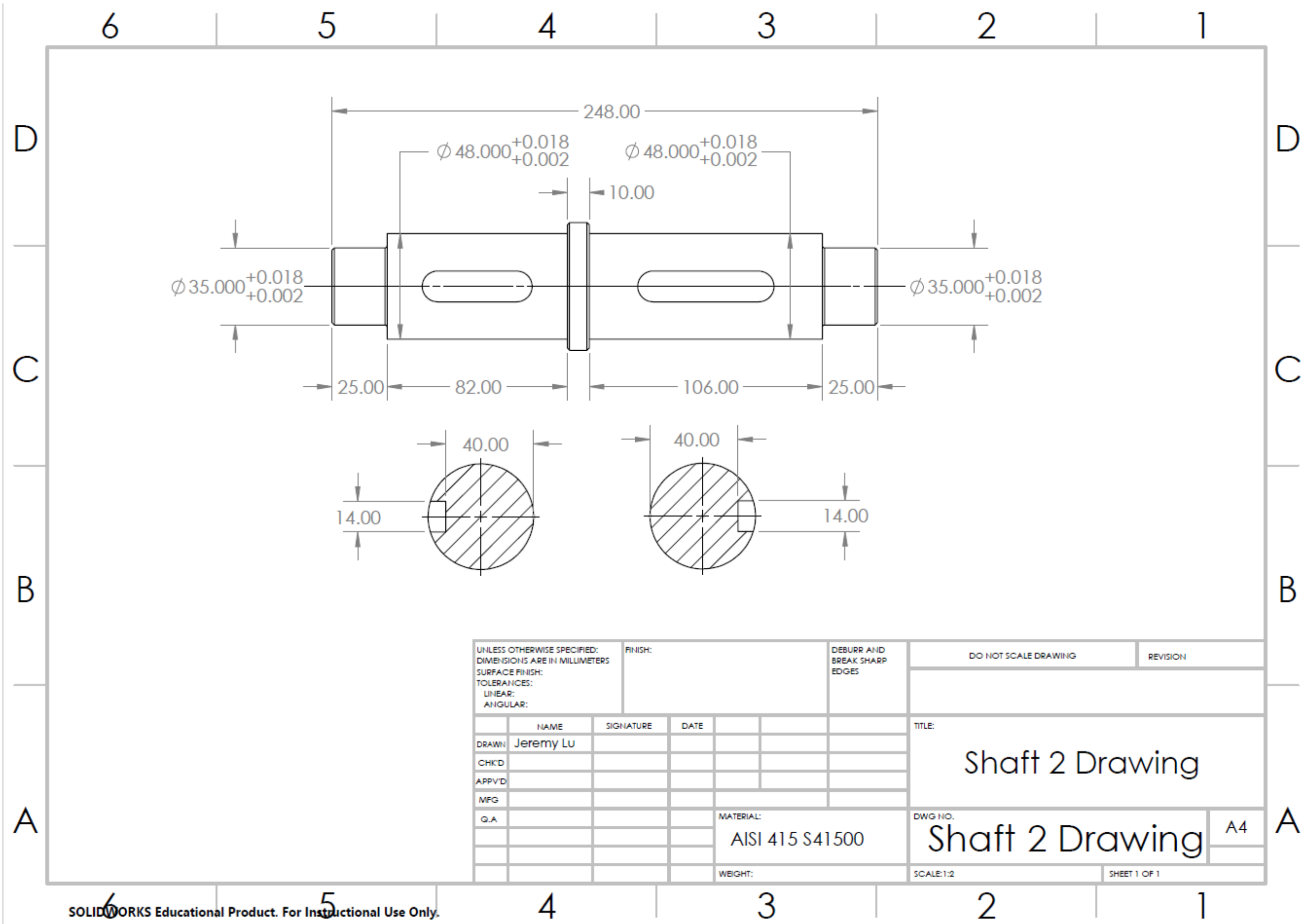

Figure 5: Shaft 2 Drawing

6.3 Shaft 3

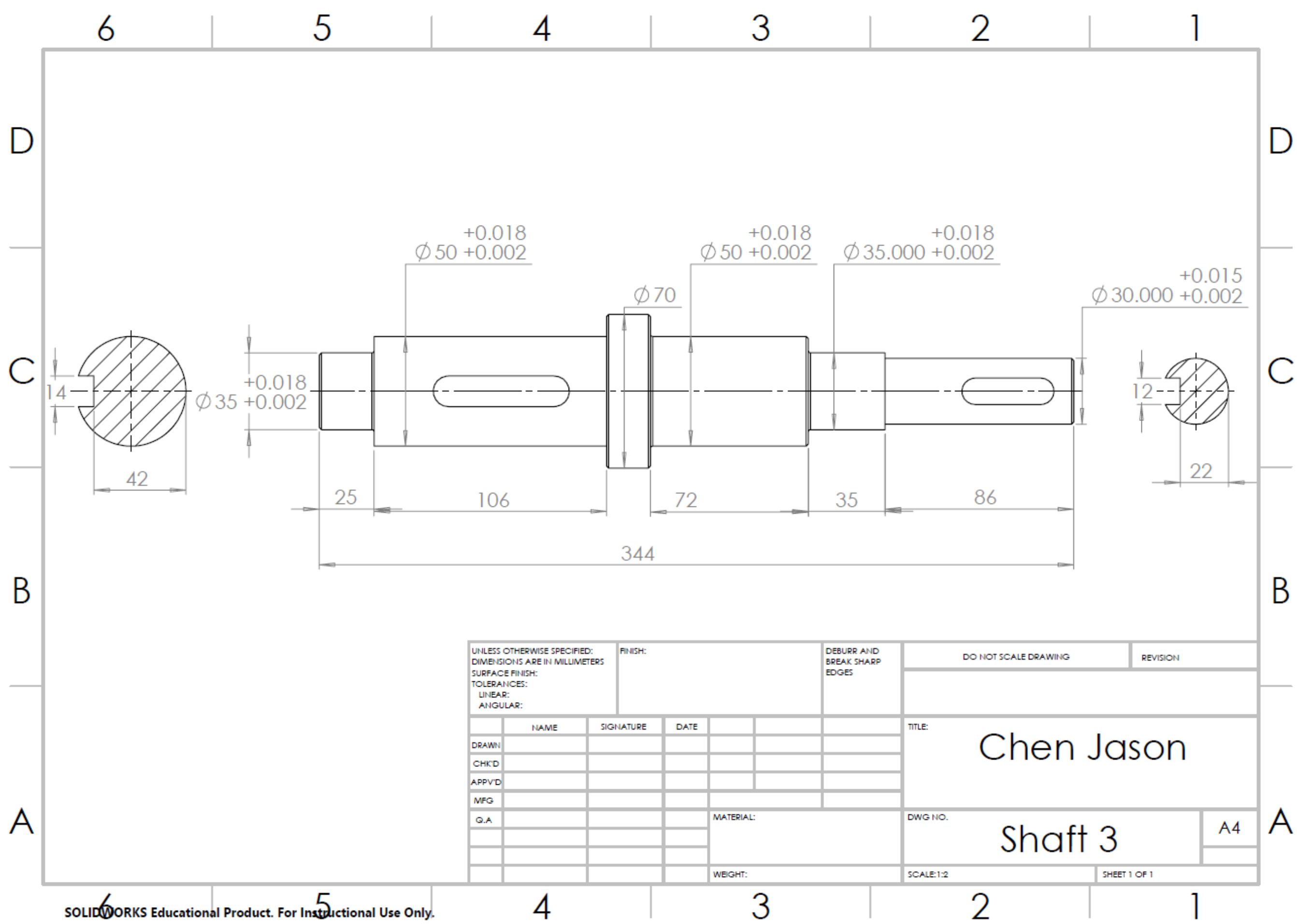

Figure 6: Shaft 3 Drawing

7 Gear Drawings

7.1 Gear 1

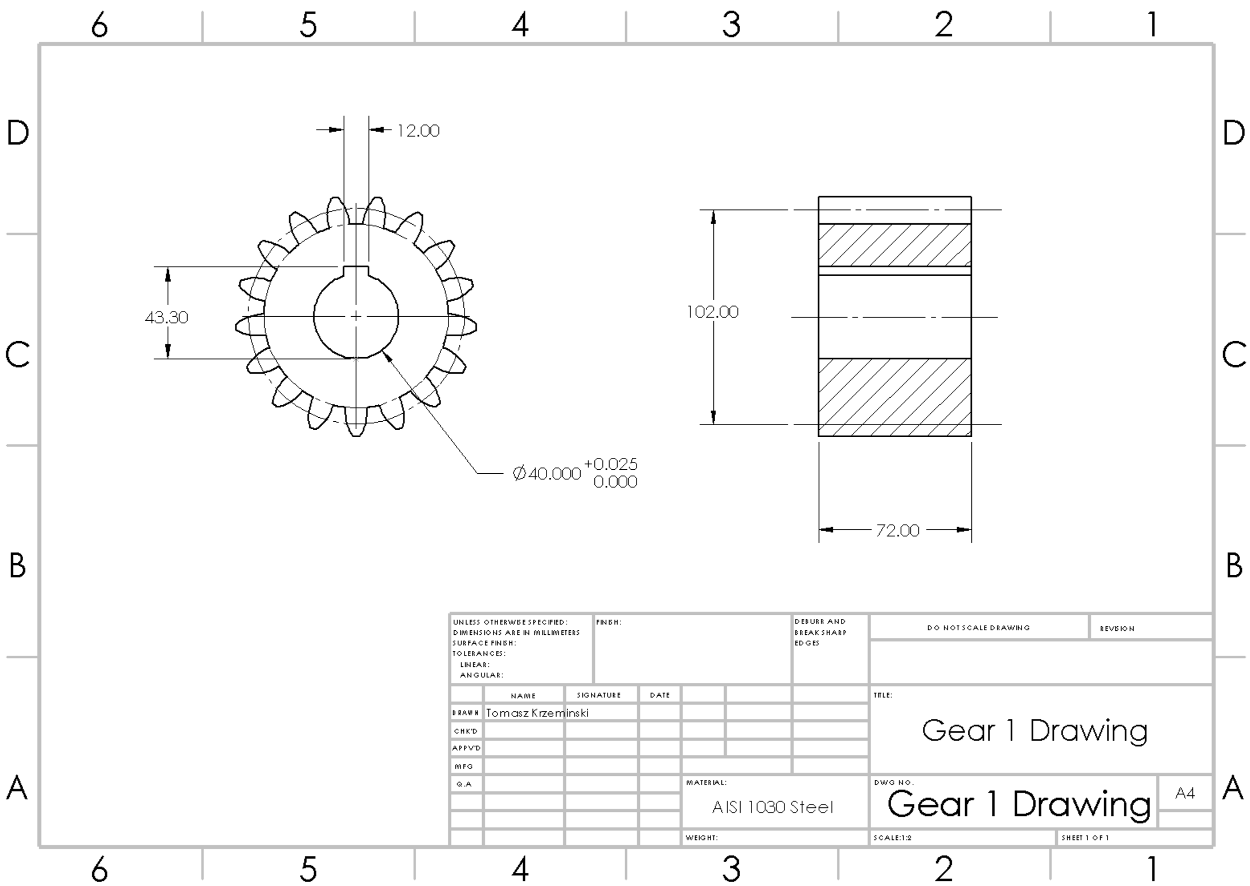

Figure 7: Gear 1 Drawing

7.2 Gear 2

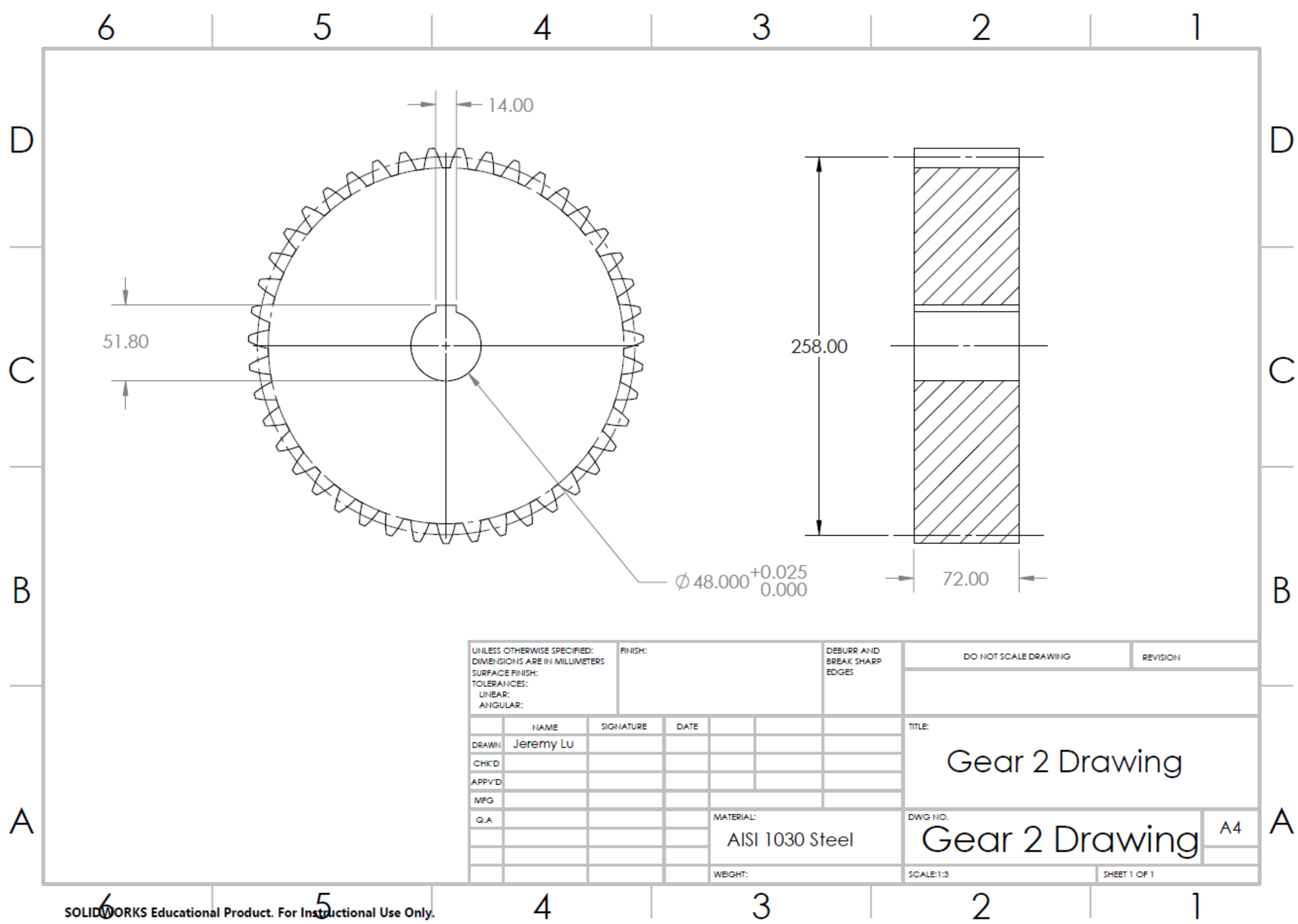

Figure 8: Gear 2 Drawing

7.3 Gear 3

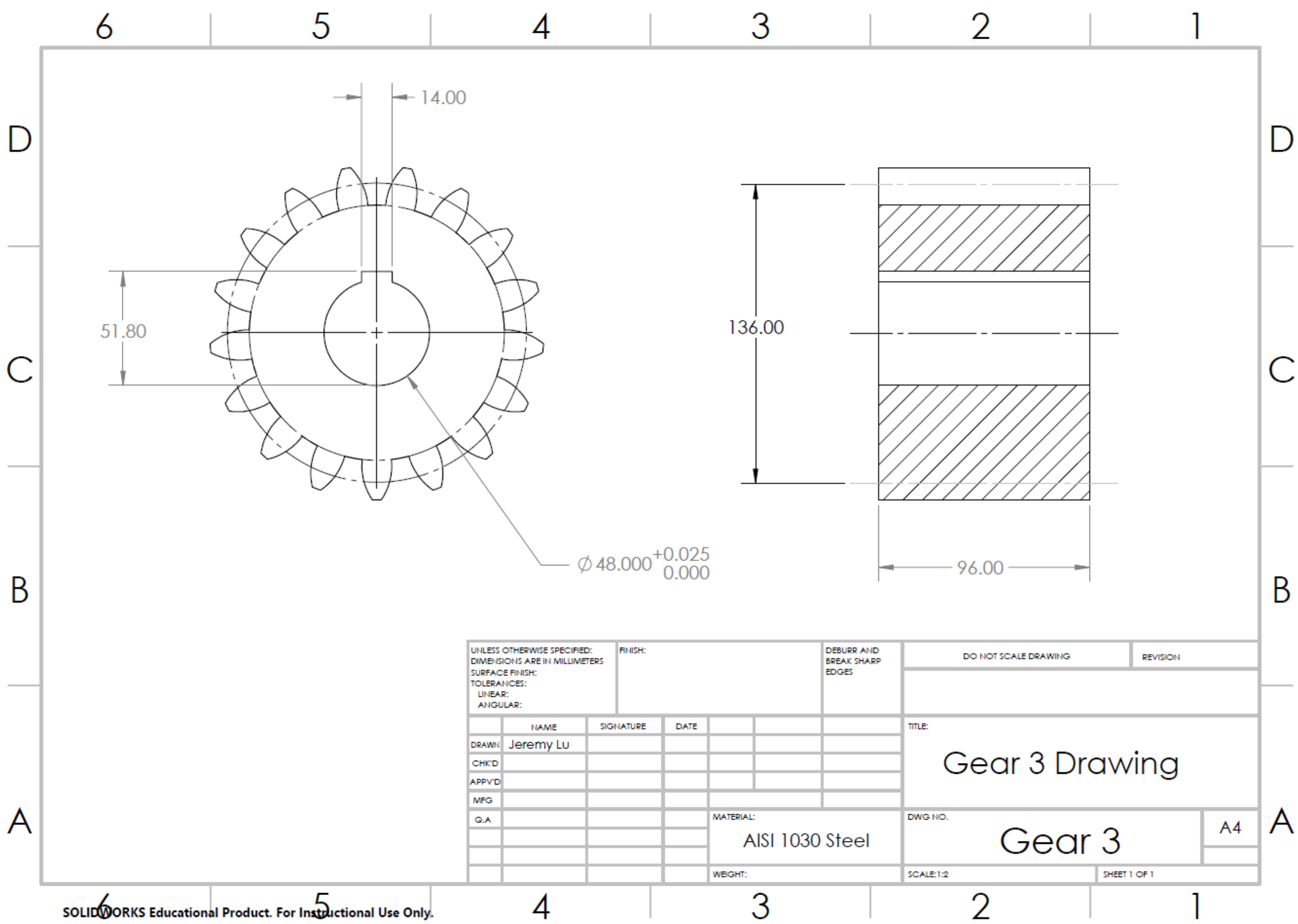

Figure 9: Gear 3 Drawing

7.4 Gear 4

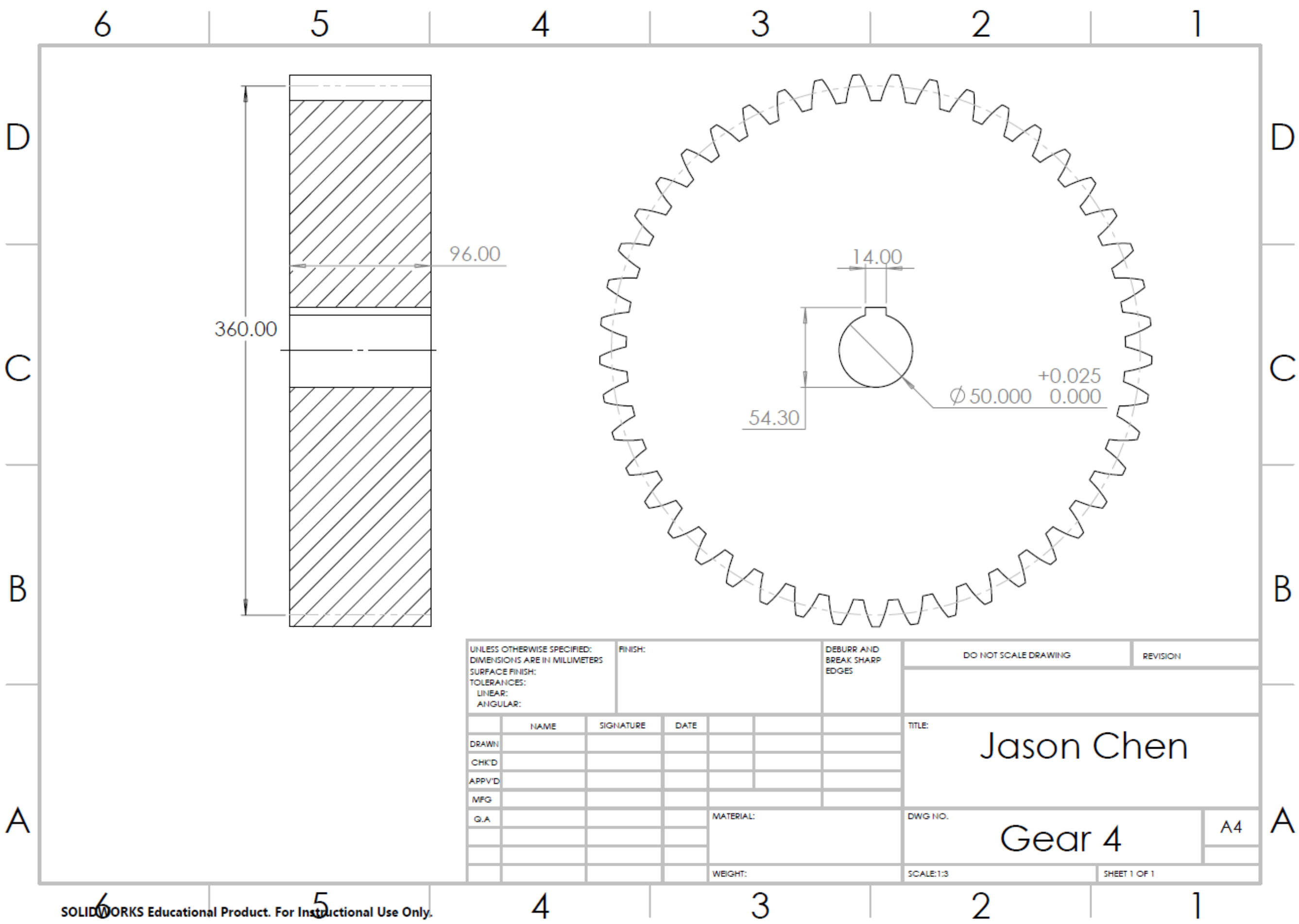

Figure 10: Gear 4 Drawing

A Shaft 1 Calculations, Tomasz Krzeminski

Statics for Shaft 1

Equilibrium Equations for Shaft 1

Torque

$$T_1 = \frac{60H}{2\pi 2700} = 212\text{Nm}$$

Tangential Forces

$$W_{\mathrm{t}I} = \frac{T_1}{\frac{270}{2}} = 393\text{N}$$

$$W_{\mathrm{t}1} = \frac{T_1}{\frac{102}{2}} = -4161\text{N}$$

Radial Forces

$$W_{\mathrm{r}I} = Wt_I \times \tan(20) = 143\text{N}$$

$$W_{\mathrm{r}1} = Wt_1 \times \tan(20) = 1514\text{N}$$

$$\sum F_z = 0 = A_z + B_z + W_{t1} + W_{tI} \rightarrow A_z = \mathbf{2644\ N}$$
$$\sum F_y = 0 = A_y + B_y + W_{r1} + W_{rI} \rightarrow A_y = \mathbf{962\ N}$$
$$\sum M_y = 0 = 237.0B_z + 58.5W_{t1} - 58.5W_{tI} \rightarrow B_z = \mathbf{1124\ N}$$
$$\sum M_z = 0 = 237.0B_y + 58.5W_{r1} - 58.5W_{rI} \rightarrow B_y = \mathbf{409\ N}$$

Maximum Bending Moment on Shaft 1

$$M_{\max} = \sqrt{M_{\mathrm{z1}}^2 + M_{\mathrm{y1}}^2} = \mathbf{213\ Nm}$$

Maximum Shear Stress on Shaft 1

$$V_{\max} = \sqrt{V_{\mathrm{z1}}^2 + V_{\mathrm{y1}}^2} = \mathbf{3232\ N}$$

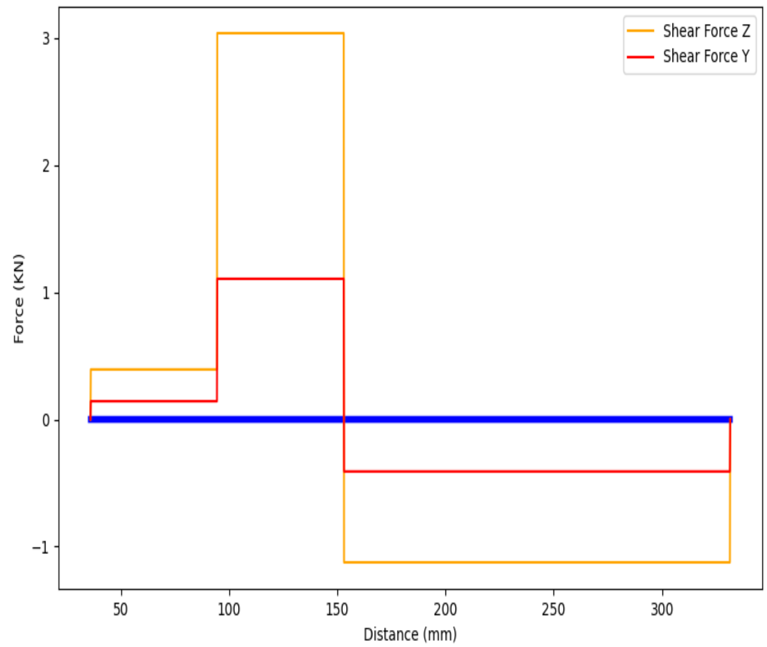

Figure 11: Shear Force Diagram Shaft 1

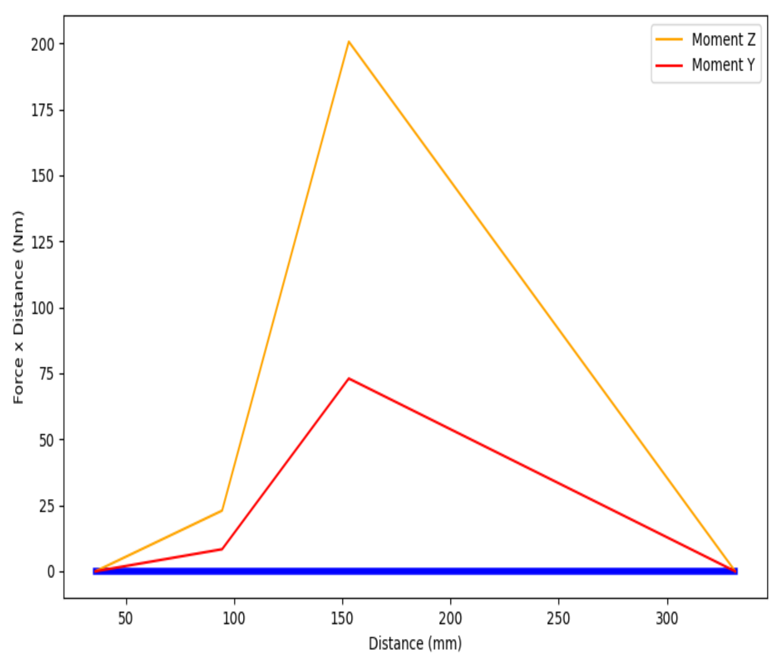

Figure 12: Moment Diagram Shaft 1

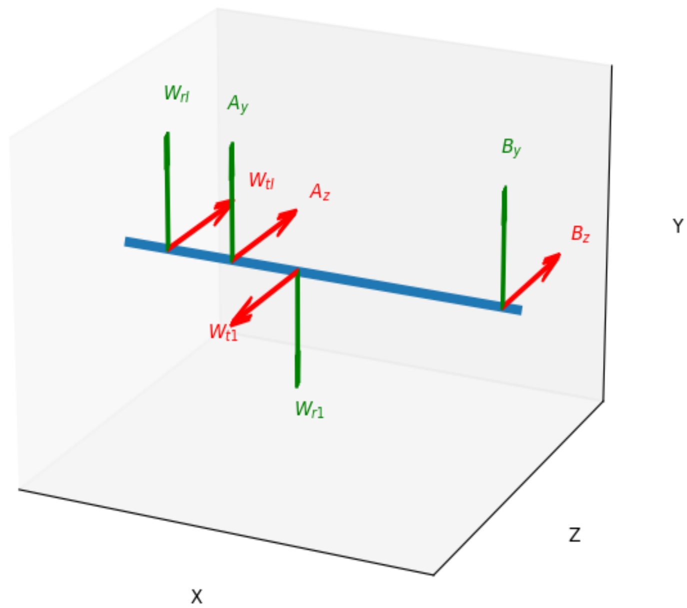

Figure 13: Free Body Diagram Shaft 1

Shaft Analysis for Shaft 1

Minimum Diameter

$$d_{\text{min}} = \sqrt[3]{\frac{16T}{2\pi\tau_{\text{all}}}} = \mathbf{15.03\ mm}$$

Account for Keyway by %5 Increase

$$d'_{\text{min}} = 1.05 \times d_{\text{min}} = \mathbf{15.79\ mm}$$

Because d_{min} is less than 20 mm:

$$d_{\text{min}} = \mathbf{20.00\ mm}$$

For Critical Cross Section Choose:

$$d = \mathbf{40\ mm}$$

Stresses

$$\sigma_m = \frac{F_x}{A} = \mathbf{0\ MPa}$$

$$\sigma_a = \frac{32M_{\max}}{\pi d^3} = \mathbf{33.97\ MPa}$$

$$\tau_m = \frac{16T}{\pi d^3} = \mathbf{16.89\ MPa}$$

$$\tau_a = \frac{16V}{3\pi d^2} = \mathbf{3.43\ MPa}$$

Concentration Factors

Kc and Kcs assumed for 1mm fillet:

$$K_c = 2.1, \quad K_{cs} = 1.5$$

$$K_f = 1 + 0.8(K_c - 1) = 1.88, \quad K_{fs} = 1 + 0.8(K_{cs} - 1) = 1.40$$

Strength Factors

$$k_t = 1.00$$
$$k_m = 1.00$$
$$k_s = 1.189d^{-0.112} = 0.79$$
$$k_r = 0.82$$
$$k_f = 0.74$$

Von Mises Stresses

$$\sigma'_{\text{Von-a}} = \sqrt{(K_f\sigma_a)^2 + 3(K_{fs}\tau_a)^2} = \mathbf{40.95\ MPa}$$

$$\sigma_{\text{Von-a}} = \sqrt{(\sigma_a)^2 + 3(\tau_a)^2} = \mathbf{29.25\ MPa}$$

$$\sigma_{\text{Von-m}} = \sqrt{(\sigma_m)^2 + 3(\tau_m)^2} = \mathbf{53.21\ MPa}$$

Fatigue Strength

$$S_{\text{Ut}} = 900\,\text{MPa}$$

$$S_{\text{y}} = 700\,\text{MPa}$$

$$S'_e = 0.5S_{\text{Ut}} = 450\,\text{MPa}$$

$$S'_l = 0.9S_{\text{Ut}} = 810\,\text{MPa}$$

$$b = \frac{-1}{3} log_1 0 \frac{S_l'}{S_e'} = -0.09$$

$$c = \log_{10}(\frac{S_l'^2}{S_e'}) = 3.16$$

$$N_c = N_c = 5 Years \times 365 \frac{Days}{Year} \times 8 \frac{Hours}{Day} 60 \frac{Min}{Hours} \times 2700 Rpm = 2365200000$$

$$S_f' = 10^c N^b = 232.34\,\mathrm{MPa}$$

$$S_f = k_f k_s k_r k_t k_m S_e' = \mathbf{111.43\ MPa}$$

Goodman

$$n_s = \frac{1}{\frac{\sigma'_{\text{Von-a}}}{S_f} + \frac{\sigma_{\text{Von-m}}}{S_{\text{Ut}}}} = \mathbf{1.45}$$

Yield

$$n_s = \frac{1}{\frac{\sigma_{\text{Von-a}}}{S_y} + \frac{\sigma_{\text{Von-m}}}{S_y}} = \mathbf{14.31}$$

Gear Analysis for Shaft 1 Gear 1

Factors for Allowable Bending

$$N_c = 5 Years \times 365 \frac{Days}{Year} \times 8 \frac{Hours}{Day} 60 \frac{Min}{Hours} \times 2700 Rpm = 2365\,\text{Millions of Cycles}$$

$$Y_n = 1.683 N_c^{\text{-0.0323}} = 0.84$$

$$S_b = 0.703 HB_p + 113 = 394.90\,\mathrm{MPa}$$

$$k_t = 1.00$$

$$k_r = 1.00$$

Allowable Bending Stress

$$\sigma_{\text{all-b}} = \frac{S_b Y_n}{k_t k_r} = \mathbf{331.00\ MPa}$$

Factors for Normal Bending Stress

Load Distribution Factor

$$N = 17$$
$$m = 6$$
$$b_w = 72$$
$$C_{\text{pf}} = \frac{b_w}{10Nm} - 0.0375 + 0.000492 b_w = 0.07$$
$$C_{\text{pm}} = 1.1$$
$$C_{\text{ma}} = 0.127 + 0.000622 \times b_w - 1.69 \times 10^{\text{-7}} * b_w^2 = 0.17$$
$$C_e = 1$$
$$C_{\text{mc}} = 1$$

Dynamic Factor

$$B = \frac{(12-6)^{\frac{3}{2}}}{4} = 0.83$$
$$A = 50 + 56(1-B) = 59.77$$
$$V = 2\pi \frac{N \times m \times 0.001}{2} \times \frac{n}{60} = 14.42$$

Factors for Normal Bending Stress

$$Y_j = 0.32$$
$$k_a = 1.00$$
$$k_s = 1.05$$
$$k_m = 1 + C_{\text{mc}}(C_{\text{mpf}} \times C_{\text{pm}} + C_{\text{ma}} \times C_e) = 1.25$$
$$k_i = 1$$
$$k_b = 1$$
$$k_v = (\frac{A + \sqrt{200V}}{A})^B = 1.70$$

Normal Bending Stress

$$\sigma_b = \frac{W_t}{m b_w Y_j} k_a k_s k_m k_v k_i k_b = \mathbf{66.86\ MPa}$$

Factors for Allowable Contact

$$N_c = 5 Years \times 365 \frac{Days}{Year} \times 8 \frac{Hours}{Day} 60 \frac{Min}{Hours} \times 2700 Rpm = 2365\,\text{Millions of Cycles}$$
$$Z_n = 2.466 N_c^{\text{-0.056}} = 0.74$$
$$S_c = 2.41 HB_p + 237 = 1203.41\,\text{MPa}$$
$$C_{\text{Hg}} = 1.00 = 1.00$$
$$k_t = 1.00$$
$$k_r = 1.00$$

Allowable Contact Stress

$$\sigma_{\text{all-c}} = \frac{S_c Z_n C_{\text{Hg}}}{k_t k_r} = \mathbf{886.07\ MPa}$$

Factors for Normal Contact Stress

Factors for Normal Contact Stress

$$K_E = \sqrt{\frac{2}{\frac{1-v_p^2}{E} + \frac{1-v_p^2}{E}}} = 467294.82$$

$$I = \frac{\pi \cos(\phi)\sin(\phi)}{1 + \frac{d_p}{d_g}} = 0.72$$

$$k_a = 1.00$$

$$k_s = 1.05$$

$$k_m = 1 + C_{\text{mc}}(C_{\text{mpf}} \times C_{\text{pm}} + C_{\text{ma}} \times C_e) = 1.25$$

$$k_v = (\frac{A + \sqrt{200V}}{A})^B = 1.70$$

Normal Contact Stress

$$\sigma_c = K_E \sqrt{\frac{W_t}{d_p b_w I} k_a k_s k_m k_v} = \mathbf{616.27\ MPa}$$

Bending Factor of Safety

$$n_b = \frac{\sigma_{\text{all-b}}}{\sigma_b} = \mathbf{4.95}$$

Contact Factor of Safety

$$n_c = \frac{\sigma_{\text{all-c}}}{\sigma_c} = \mathbf{1.44}$$

Bearing Analysis for Shaft 1

Life Cycle of Bearing

$$h = 5\ Years \times 52 \frac{Weeks}{Year} \times 7 \frac{Days}{Week} \times 8 \frac{Hours}{Day} = 14560 \text{ Hours}$$

$$n_1 = 2700 \text{ Rpm}$$

$$L_{10} = 60 \times h \times n_1 = 2359 \text{ Millions of Cycles}$$

Critical Bearing Force

$$A_p = 1.0 \text{ No Shock Present}$$

$$m = 3 \text{ Ball Bearing}$$

$$P_r = 2814 \text{ N}$$

$$X = 1.00 \text{ No Axial Force}$$

$$P = P \times X = \mathbf{2814\ N}$$

Left Bearing C

$$C_l = PL_{10}^{\frac{1}{3}} = \mathbf{37452\ N}$$

Right Bearing C

$$C_r = PL_{10}^{\frac{1}{3}} = \mathbf{15923\ N}$$

Bearing Selection

$$\text{Choose SKF } \mathbf{6307} : C = \mathbf{55300\ N}$$

Same Bearing Used for Both Sides

Left Bearing FOS

$$n_s = \frac{55300}{37452} = \mathbf{1.09}$$

Right Bearing FOS

$$n_s = \frac{55300}{15923} = \mathbf{2.57}$$

B Shaft 2 Calculations, Jeremy Lu

Statics for Shaft 2

Equilibrium Equations for Shaft 2

Torque

$$T_2 = W_{t2} \times \frac{258}{2} = 537\text{Nm}$$

Tangential Forces

$$W_{\text{t2}} = \frac{T_2}{\frac{258}{2}} = 4161\text{N}$$

$$W_{\text{t3}} = \frac{T_2}{\frac{136}{2}} = -7893\text{N}$$

Radial Forces

$$W_{\text{r2}} = Wt_2 \times \tan(20) = 1514\text{N}$$

$$W_{\text{r3}} = Wt_3 \times \tan(20) = -2873\text{N}$$

$$\sum F_z = 0 = C_z + D_z + W_{t2} + W_{t3} \rightarrow C_z = \textbf{-574 N}$$
$$\sum F_y = 0 = C_y + D_y + W_{r2} + W_{r3} \rightarrow C_y = \textbf{-209 N}$$
$$\sum M_y = 0 = 223.0D_z + 58.5W_{t2} + 152.5W_{t3} \rightarrow D_z = \textbf{4306 N}$$
$$\sum M_z = 0 = 223.0D_y + 58.5W_{r2} + 152.5W_{r3} \rightarrow D_y = \textbf{1567 N}$$

Maximum Bending Moment on Shaft 2

$$M_{\text{max}} = \sqrt{M_{\text{z2}}^2 + M_{\text{y2}}^2} = \textbf{323 Nm}$$

Maximum Shear Stress on Shaft 2

$$V_{\text{max}} = \sqrt{V_{\text{z2}}^2 + V_{\text{y2}}^2} = \textbf{4583 N}$$

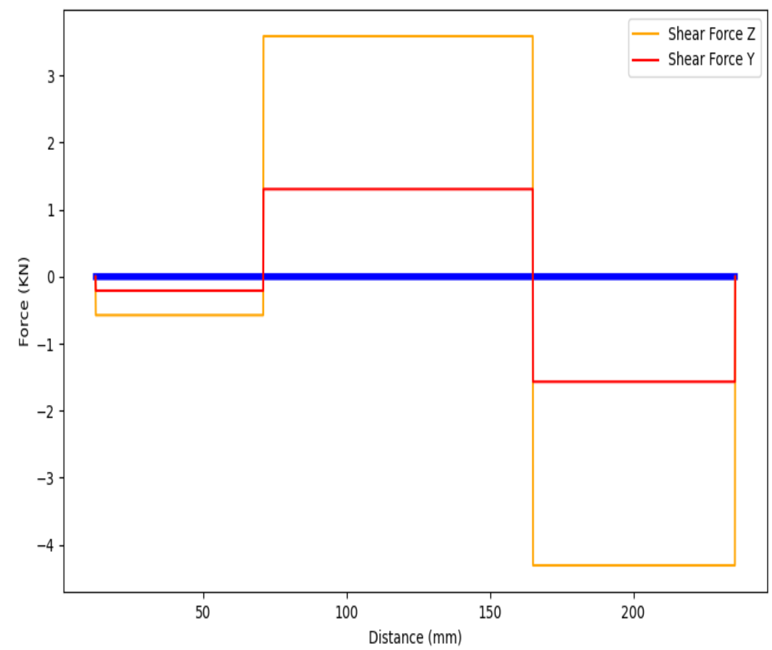

Figure 14: Shear Force Diagram Shaft 2

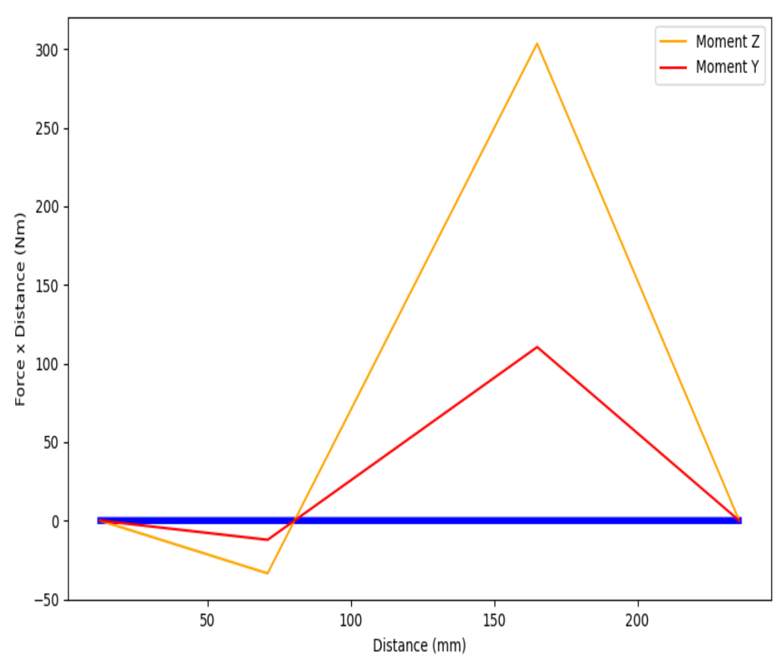

Figure 15: Moment Diagram Shaft 2

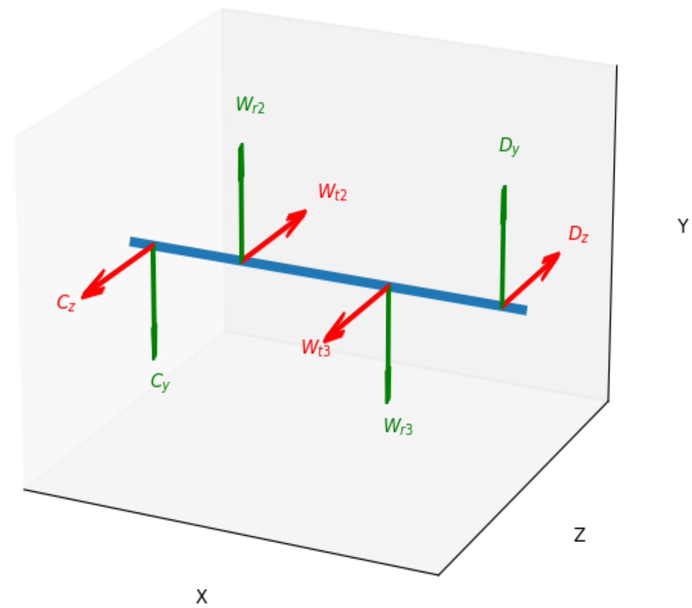

Figure 16: Free Body Diagram Shaft 2

Shaft Analysis for Shaft 2

Minimum Diameter

$$d_{\min} = \sqrt[3]{\frac{16T}{2\pi\tau_{\text{all}}}} = \mathbf{20.48\ mm}$$

Account for Keyway by %5 Increase

$$d'_{\min} = 1.05 \times d_{\min} = \mathbf{21.51\ mm}$$

Because $d_{\min}$ is greater than 20 mm:

$$d_{\min} = \mathbf{21.51\ mm}$$

For Critical Cross Section Choose:

$$d = \mathbf{48\ mm}$$

Stresses

$$\sigma_m = \frac{F_x}{A} = \mathbf{0\ MPa}$$

$$\sigma_a = \frac{32M_{\text{max}}}{\pi d^3} = \mathbf{29.75\ MPa}$$

$$\tau_m = \frac{16T}{\pi d^3} = \mathbf{24.72\ MPa}$$

$$\tau_a = \frac{16V}{3\pi d^2} = \mathbf{3.38\ MPa}$$

Concentration Factors

Kc and Kcs assumed for 1mm fillet:

$$K_c = 2.1, \quad K_{cs} = 1.5$$

$$K_f = 1 + 0.8(K_c - 1) = 1.88, \quad K_{fs} = 1 + 0.8(K_{cs} - 1) = 1.40$$

Strength Factors

$$k_t = 1.00$$
$$k_m = 1.00$$
$$k_s = 1.189d^{-0.112} = 0.77$$
$$k_r = 0.82$$
$$k_f = 0.74$$

Von Mises Stresses

$$\sigma'_{\text{Von-a}} = \sqrt{(K_f\sigma_a)^2 + 3(K_{fs}\tau_a)^2} = \mathbf{59.94\ MPa}$$

$$\sigma_{\text{Von-a}} = \sqrt{(\sigma_a)^2 + 3(\tau_a)^2} = \mathbf{42.81\ MPa}$$

$$\sigma_{\text{Von-m}} = \sqrt{(\sigma_m)^2 + 3(\tau_m)^2} = \mathbf{66.91\ MPa}$$

Fatigue Strength

$$S_{\text{Ut}} = 900\,\text{MPa}$$

$$S_{\text{y}} = 700\,\text{MPa}$$

$$S'_e = 0.5S_{\text{Ut}} = 450\,\text{MPa}$$

$$S'_l = 0.9S_{\text{Ut}} = 810\,\text{MPa}$$

$$b = \frac{-1}{3} log_1 0 \frac{S_l'}{S_e'} = -0.09$$

$$c = \log_{10}(\frac{S_l'^2}{S_e'}) = 3.16$$

$$N_c = N_c = 5 Years \times 365 \frac{Days}{Year} \times 8 \frac{Hours}{Day} 60 \frac{Min}{Hours} \times 1067 Rpm = 935079070$$

$$S_f' = 10^c N^b = 251.43\,\mathrm{MPa}$$

$$S_f = k_f k_s k_r k_t k_m S_e' = \mathbf{118.15\ MPa}$$

Goodman

$$n_s = \frac{1}{\frac{\sigma'_{\text{Von-a}}}{S_f} + \frac{\sigma_{\text{Von-m}}}{S_{\text{Ut}}}} = \mathbf{1.43}$$

Yield

$$n_s = \frac{1}{\frac{\sigma_{\text{Von-a}}}{S_y} + \frac{\sigma_{\text{Von-m}}}{S_y}} = \mathbf{12.27}$$

Gear Analysis for Shaft 2 Gear 2

Factors for Allowable Bending

$$N_c = 5 Years \times 365 \frac{Days}{Year} \times 8 \frac{Hours}{Day} 60 \frac{Min}{Hours} \times 1067 Rpm = 935\,\text{Millions of Cycles}$$

$$Y_n = 1.683 N_c^{-0.0323} = 0.86$$

$$S_b = 0.533 HB_g + 88.3 = 249.27\,\mathrm{MPa}$$

$$k_t = 1.00$$

$$k_r = 1.00$$

Allowable Bending Stress

$$\sigma_{\text{all-b}} = \frac{S_b Y_n}{k_t k_r} = \mathbf{215.29\ MPa}$$

Factors for Normal Bending Stress

Load Distribution Factor

$$N = 43$$
$$m = 6$$
$$b_w = 72$$
$$C_{\text{pf}} = \frac{b_w}{10Nm} - 0.0375 + 0.000492 b_w = 0.03$$
$$C_{\text{pm}} = 1.1$$
$$C_{\text{ma}} = 0.127 + 0.000622 \times b_w - 1.69 \times 10^{\text{-7}} * b_w^2 = 0.17$$
$$C_e = 1$$
$$C_{\text{mc}} = 1$$

Dynamic Factor

$$B = \frac{(12-6)^{\frac{3}{2}}}{4} = 0.83$$
$$A = 50 + 56(1 - B) = 59.77$$
$$V = 2\pi \frac{N \times m \times 0.001}{2} \times \frac{n}{60} = 14.42$$

Factors for Normal Bending Stress

$$Y_j = 0.37$$
$$k_a = 1.00$$
$$k_s = 1.05$$
$$k_m = 1 + C_{\text{mc}}(C_{\text{mpf}} \times C_{\text{pm}} + C_{\text{ma}} \times C_e) = 1.25$$
$$k_i = 1$$
$$k_b = 1$$
$$k_v = (\frac{A + \sqrt{200V}}{A})^B = 1.70$$

Normal Bending Stress

$$\sigma_b = \frac{W_t}{m b_w Y_j} k_a k_s k_m k_v k_i k_b = \mathbf{57.83\ MPa}$$

Factors for Allowable Contact

$$N_c = 5Years \times 365\frac{Days}{Year} \times 8\frac{Hours}{Day}60\frac{Min}{Hours} \times 1067Rpm = 935\,\text{Millions of Cycles}$$

$$Z_n = 2.466N_c^{\text{-0.056}} = 0.78$$

$$S_c = 2.22HB_g + 200 = 870.44\,\text{MPa}$$

$$C_{\text{Hg}} = A_p\left(\frac{N_g}{N_p} - 1\right) + 1 = 1.01$$

$$k_t = 1.00$$

$$k_r = 1.00$$

Allowable Contact Stress

$$\sigma_{\text{all-c}} = \frac{S_c Z_n C_{\text{Hg}}}{k_t k_r} = \mathbf{678.72\ MPa}$$

Factors for Normal Contact Stress

Factors for Normal Contact Stress

$$K_E = \sqrt{\frac{2}{\frac{1-v_p^2}{E} + \frac{1-v_p^2}{E}}} = 467294.82$$

$$I = \frac{\pi\cos(\phi)\sin(\phi)}{1 + \frac{d_p}{d_g}} = 0.72$$

$$k_a = 1.00$$

$$k_s = 1.05$$

$$k_m = 1 + C_{\text{mc}}(C_{\text{mpf}} \times C_{\text{pm}} + C_{\text{ma}} \times C_e) = 1.25$$

$$k_v = (\frac{A + \sqrt{200V}}{A})^B = 1.70$$

Normal Contact Stress

$$\sigma_c = K_E\sqrt{\frac{W_t}{d_p b_w I}k_a k_s k_m k_v} = \mathbf{616.27\ MPa}$$

Bending Factor of Safety

$$n_b = \frac{\sigma_{\text{all-b}}}{\sigma_b} = \mathbf{3.72}$$

Contact Factor of Safety

$$n_c = \frac{\sigma_{\text{all-c}}}{\sigma_c} = \mathbf{1.10}$$

Gear Analysis for Shaft 2 Gear 3

Factors for Allowable Bending

$$N_c = 5Years \times 365\frac{Days}{Year} \times 8\frac{Hours}{Day}60\frac{Min}{Hours} \times 1067Rpm = 935\,\text{Millions of Cycles}$$
$$Y_n = 1.683N_c^{-0.0323} = 0.86$$
$$S_b = 0.703HB_p + 113 = 394.90\,\text{MPa}$$
$$k_t = 1.00$$
$$k_r = 1.00$$

Allowable Bending Stress

$$\sigma_{\text{all-b}} = \frac{S_b Y_n}{k_t k_r} = \textbf{341.07 MPa}$$

Factors for Normal Bending Stress

Load Distribution Factor

$$N = 17$$
$$m = 8$$
$$b_w = 96$$
$$C_{\text{pf}} = \frac{b_w}{10Nm} - 0.0375 + 0.000492b_w = 0.08$$
$$C_{\text{pm}} = 1.1$$
$$C_{\text{ma}} = 0.127 + 0.000622 \times b_w - 1.69 \times 10^{-7} * b_w^2 = 0.19$$
$$C_e = 1$$
$$C_{\text{mc}} = 1$$

Dynamic Factor

$$B = \frac{(12-6)^{\frac{3}{2}}}{4} = 0.83$$
$$A = 50 + 56(1 - B) = 59.77$$
$$V = 2\pi\frac{N \times m \times 0.001}{2} \times \frac{n}{60} = 7.60$$

Factors for Normal Bending Stress

$$Y_j = 0.32$$
$$k_a = 1.00$$
$$k_s = 1.05$$
$$k_m = 1 + C_{\text{mc}}(C_{\text{mpf}} \times C_{\text{pm}} + C_{\text{ma}} \times C_e) = 1.27$$
$$k_i = 1$$
$$k_b = 1$$
$$k_v = (\frac{A + \sqrt{200V}}{A})^B = 1.51$$

Normal Bending Stress

$$\sigma_b = \frac{W_t}{mb_w Y_j} k_a k_s k_m k_v k_i k_b = \mathbf{65.01\ MPa}$$

Factors for Allowable Contact

$$N_c = 5 Years \times 365 \frac{Days}{Year} \times 8 \frac{Hours}{Day} 60 \frac{Min}{Hours} \times 1067 Rpm = 935\,\text{Millions of Cycles}$$
$$Z_n = 2.466 N_c^{\text{-0.056}} = 0.78$$
$$S_c = 2.41 HB_p + 237 = 1203.41\,\text{MPa}$$
$$C_{\text{Hg}} = 1.00 = 1.00$$
$$k_t = 1.00$$
$$k_r = 1.00$$

Allowable Contact Stress

$$\sigma_{\text{all-c}} = \frac{S_c Z_n C_{\text{Hg}}}{k_t k_r} = \mathbf{933.34\ MPa}$$

Factors for Normal Contact Stress

Factors for Normal Contact Stress

$$K_E = \sqrt{\frac{2}{\frac{1-v_p^2}{E} + \frac{1-v_p^2}{E}}} = 467294.82$$
$$I = \frac{\pi \cos(\phi) \sin(\phi)}{1 + \frac{d_p}{d_g}} = 0.73$$
$$k_a = 1.00$$
$$k_s = 1.05$$
$$k_m = 1 + C_{\text{mc}}(C_{\text{mpf}} \times C_{\text{pm}} + C_{\text{ma}} \times C_e) = 1.27$$
$$k_v = (\frac{A + \sqrt{200V}}{A})^B = 1.51$$

Normal Contact Stress

$$\sigma_c = K_E \sqrt{\frac{W_t}{d_p b_w I} k_a k_s k_m k_v} = \mathbf{603.85\ MPa}$$

Bending Factor of Safety

$$n_b = \frac{\sigma_{\text{all-b}}}{\sigma_b} = \mathbf{5.25}$$

Contact Factor of Safety

$$n_c = \frac{\sigma_{\text{all-c}}}{\sigma_c} = \mathbf{1.55}$$

Bearing Analysis for Shaft 2

Life Cycle of Bearing

$$h = 5\ Years \times 52 \frac{Weeks}{Year} \times 7 \frac{Days}{Week} \times 8 \frac{Hours}{Day} = 14560 \text{ Hours}$$

$$n_2 = 1067 \text{ Rpm}$$

$$L_{10} = 60 \times h \times n_2 = 933\,\text{Millions of Cycles}$$

Critical Bearing Force

$$A_p = 1.0 \text{ No Shock Present}$$

$$m = 3 \text{ Ball Bearing}$$

$$P_r = 4583 \text{ N}$$

$$X = 1.00 \text{ No Axial Force}$$

$$P = P \times X = \mathbf{4583\ N}$$

Left Bearing C

$$C_l = PL_{10}^{\frac{1}{3}} = \mathbf{5967\ N}$$

Right Bearing C

$$C_r = PL_{10}^{\frac{1}{3}} = \mathbf{44773\ N}$$

Bearing Selection

$$\text{Choose SKF } \mathbf{6407} : C = \mathbf{55300\ N}$$

Same Bearing Used for Both Sides

Left Bearing FOS

$$n_s = \frac{55300}{5967} = \mathbf{9.27}$$

Right Bearing FOS

$$n_s = \frac{55300}{44773} = \mathbf{1.24}$$

C Shaft 3 Calculations, Jason Chen

Statics for Shaft 3

Equilibrium Equations for Shaft 3

Torque

$$T_3 = W_{t4} \times \frac{360}{2} = 1421\text{Nm}$$

Tangential Forces

$$W_{\text{t}O} = \frac{T_3}{\frac{270}{2}} = -2631\text{N}$$

$$W_{\text{t4}} = \frac{T_3}{\frac{360}{2}} = 7893\text{N}$$

Radial Forces

$$W_{\text{r}O} = Wt_O \times \tan(20) = -958\text{N}$$

$$W_{\text{r4}} = Wt_4 \times \tan(20) = 2873\text{N}$$

$$\sum F_z = 0 = E_z + F_z + W_{t4} + W_{tO} \rightarrow E_z = \textbf{1038 N}$$
$$\sum F_y = 0 = E_y + F_y + W_{r4} + W_{rO} \rightarrow E_y = \textbf{378 N}$$
$$\sum M_y = 0 = 213.0F_z + 142.5W_{t4} - 82.5W_{tO} \rightarrow F_z = \textbf{-6300 N}$$
$$\sum M_z = 0 = 213.0F_y + 142.5W_{r4} - 82.5W_{rO} \rightarrow F_y = \textbf{-2293 N}$$

Maximum Bending Moment on Shaft 3

$$M_{\max} = \sqrt{M_{\text{z3}}^2 + M_{\text{y3}}^2} = \textbf{473 Nm}$$

Maximum Shear Stress on Shaft 3

$$V_{\max} = \sqrt{V_{\text{z3}}^2 + V_{\text{y3}}^2} = \textbf{6704 N}$$

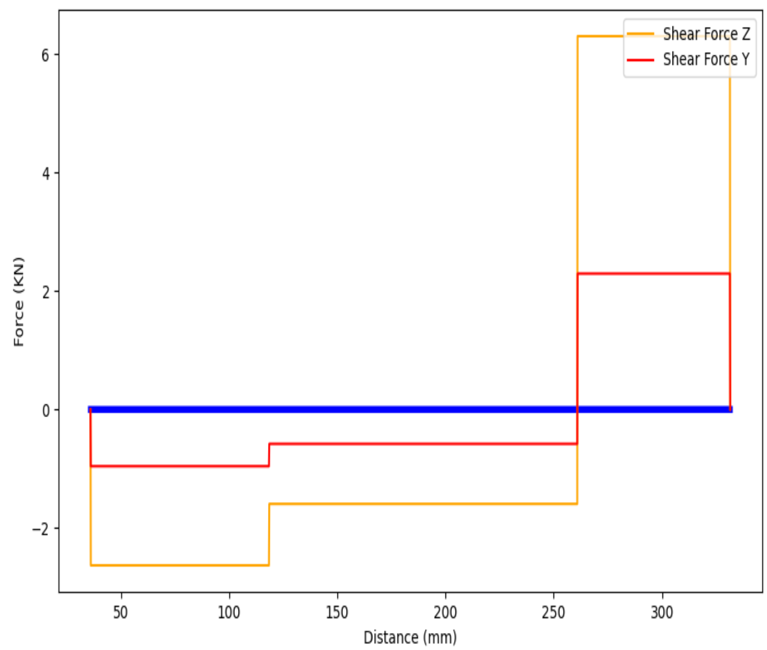

Figure 17: Shear Force Diagram Shaft 3

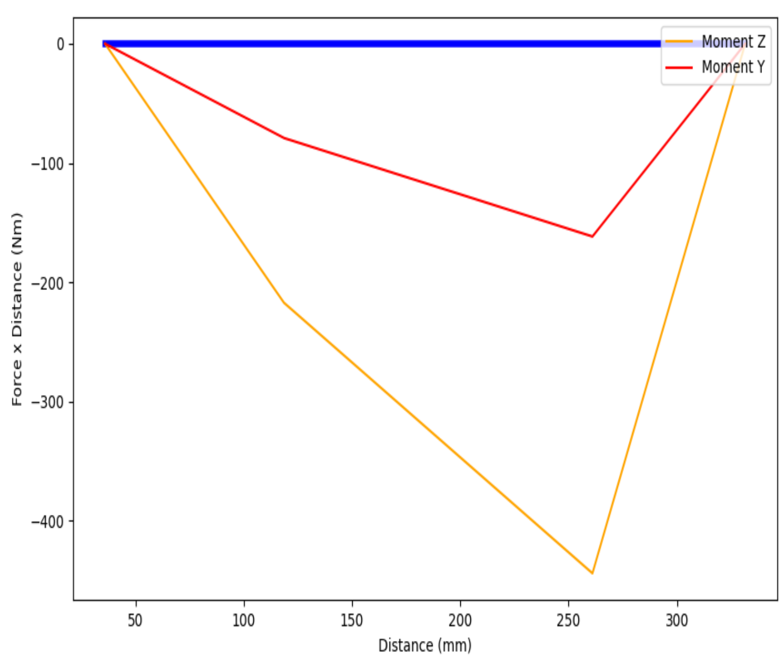

Figure 18: Moment Diagram Shaft 3

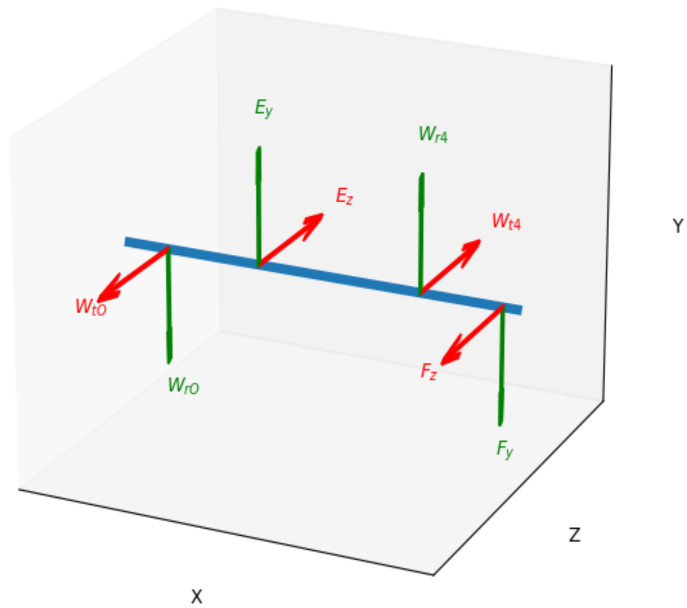

Figure 19: Free Body Diagram Shaft 3

Shaft Analysis for Shaft 3

Minimum Diameter

$$d_{\text{min}} = \sqrt[3]{\frac{16T}{2\pi\tau_{\text{all}}}} = \mathbf{28.34\ mm}$$

Account for Keyway by %5 Increase

$$d'_{\text{min}} = 1.05 \times d_{\text{min}} = \mathbf{29.75\ mm}$$

Because d_{min} is greater than 20 mm:

$$d_{\text{min}} = \mathbf{29.75\ mm}$$

For Critical Cross Section Choose:

$$d = \mathbf{50\ mm}$$

Stresses

$$\sigma_m = \frac{F_x}{A} = \mathbf{0\ MPa}$$

$$\sigma_a = \frac{32M_{\max}}{\pi d^3} = \mathbf{38.51\ MPa}$$

$$\tau_m = \frac{16T}{\pi d^3} = \mathbf{\text{-}21.87\ MPa}$$

$$\tau_a = \frac{16V}{3\pi d^2} = \mathbf{4.55\ MPa}$$

Concentration Factors

Kc and Kcs assumed for 1mm fillet:

$$K_c = 2.1, \quad K_{cs} = 1.5$$

$$K_f = 1 + 0.8(K_c - 1) = 1.88, \quad K_{fs} = 1 + 0.8(K_{cs} - 1) = 1.40$$

Strength Factors

$$k_t = 1.00$$
$$k_m = 1.00$$
$$k_s = 1.189d^{\text{-}0.112} = 0.77$$
$$k_r = 0.82$$
$$k_f = 0.74$$

Von Mises Stresses

$$\sigma'_{\text{Von-a}} = \sqrt{(K_f\sigma_a)^2 + 3(K_{fs}\tau_a)^2} = \mathbf{53.03\ MPa}$$

$$\sigma_{\text{Von-a}} = \sqrt{(\sigma_a)^2 + 3(\tau_a)^2} = \mathbf{37.88\ MPa}$$

$$\sigma_{\text{Von-m}} = \sqrt{(\sigma_m)^2 + 3(\tau_m)^2} = \mathbf{65.54\ MPa}$$

Fatigue Strength

$$S_{\text{Ut}} = 900\,\text{MPa}$$

$$S_{\text{y}} = 700\,\text{MPa}$$

$$S'_e = 0.5S_{\text{Ut}} = 450\,\text{MPa}$$

$$S'_l = 0.9S_{\text{Ut}} = 810\,\text{MPa}$$

$$b = \frac{-1}{3} log_1 0 \frac{S'_l}{S'_e} = -0.09$$

$$c = \log_{10}(\frac{S'^2_l}{S'_e}) = 3.16$$

$$N_c = N_c = 5Years \times 365 \frac{Days}{Year} \times 8 \frac{Hours}{Day} 60 \frac{Min}{Hours} \times 1067Rpm = 353252093$$

$$S'_f = 10^c N^b = 273.15\,\text{MPa}$$

$$S_f = k_f k_s k_r k_t k_m S'_e = \mathbf{127.76\ MPa}$$

Goodman

$$n_s = \frac{1}{\frac{\sigma'_{\text{Von-a}}}{S_f} + \frac{\sigma_{\text{Von-m}}}{S_{\text{Ut}}}} = \mathbf{1.41}$$

Yield

$$n_s = \frac{1}{\frac{\sigma_{\text{Von-a}}}{S_y} + \frac{\sigma_{\text{Von-m}}}{S_y}} = \mathbf{14.34}$$

Gear Analysis for Shaft 3 Gear 4

Factors for Allowable Bending

$$N_c = 5Years \times 365 \frac{Days}{Year} \times 8 \frac{Hours}{Day} 60 \frac{Min}{Hours} \times 403Rpm = 353\,\text{Millions of Cycles}$$

$$Y_n = 1.683 N_c^{\text{-0.0323}} = 0.89$$

$$S_b = 0.533 HB_g + 88.3 = 249.27\,\text{MPa}$$

$$k_t = 1.00$$

$$k_r = 1.00$$

Allowable Bending Stress

$$\sigma_{\text{all-b}} = \frac{S_b Y_n}{k_t k_r} = \mathbf{222.16\ MPa}$$

Factors for Normal Bending Stress

Load Distribution Factor

$$N = 45$$
$$m = 8$$
$$b_w = 96$$
$$C_{\text{pf}} = \frac{b_w}{10Nm} - 0.0375 + 0.000492 b_w = 0.04$$
$$C_{\text{pm}} = 1.1$$
$$C_{\text{ma}} = 0.127 + 0.000622 \times b_w - 1.69 \times 10^{-7} * b_w^2 = 0.19$$
$$C_e = 1$$
$$C_{\text{mc}} = 1$$

Dynamic Factor

$$B = \frac{(12-6)^{\frac{3}{2}}}{4} = 0.83$$
$$A = 50 + 56(1 - B) = 59.77$$
$$V = 2\pi \frac{N \times m \times 0.001}{2} \times \frac{n}{60} = 7.60$$

Factors for Normal Bending Stress

$$Y_j = 0.37$$
$$k_a = 1.00$$
$$k_s = 1.05$$
$$k_m = 1 + C_{\text{mc}}(C_{\text{mpf}} \times C_{\text{pm}} + C_{\text{ma}} \times C_e) = 1.27$$
$$k_i = 1$$
$$k_b = 1$$
$$k_v = (\frac{A + \sqrt{200V}}{A})^B = 1.51$$

Normal Bending Stress

$$\sigma_b = \frac{W_t}{m b_w Y_j} k_a k_s k_m k_v k_i k_b = \mathbf{56.22\ MPa}$$

Factors for Allowable Contact

$$N_c = 5Years \times 365\frac{Days}{Year} \times 8\frac{Hours}{Day} 60\frac{Min}{Hours} \times 403Rpm = 353\,\text{Millions of Cycles}$$

$$Z_n = 2.466N_c^{\text{-0.056}} = 0.82$$

$$S_c = 2.22HB_g + 200 = 870.44\,\text{MPa}$$

$$C_{\text{Hg}} = A_p\left(\frac{N_g}{N_p} - 1\right) + 1 = 1.01$$

$$k_t = 1.00$$

$$k_r = 1.00$$

Allowable Contact Stress

$$\sigma_{\text{all-c}} = \frac{S_c Z_n C_{\text{Hg}}}{k_t k_r} = \mathbf{717.04\ MPa}$$

Factors for Normal Contact Stress

Factors for Normal Contact Stress

$$K_E = \sqrt{\frac{2}{\frac{1-v_p^2}{E} + \frac{1-v_p^2}{E}}} = 467294.82$$

$$I = \frac{\pi\cos(\phi)\sin(\phi)}{1 + \frac{d_p}{d_g}} = 0.73$$

$$k_a = 1.00$$

$$k_s = 1.05$$

$$k_m = 1 + C_{\text{mc}}(C_{\text{mpf}} \times C_{\text{pm}} + C_{\text{ma}} \times C_e) = 1.27$$

$$k_v = (\frac{A + \sqrt{200V}}{A})^B = 1.51$$

Normal Contact Stress

$$\sigma_c = K_E\sqrt{\frac{W_t}{d_p b_w I} k_a k_s k_m k_v} = \mathbf{603.85\ MPa}$$

Bending Factor of Safety

$$n_b = \frac{\sigma_{\text{all-b}}}{\sigma_b} = \mathbf{3.95}$$

Contact Factor of Safety

$$n_c = \frac{\sigma_{\text{all-c}}}{\sigma_c} = \mathbf{1.19}$$

Bearing Analysis for Shaft 3

Life Cycle of Bearing

$$h = 5\ Years \times 52 \frac{Weeks}{Year} \times 7 \frac{Days}{Week} \times 8 \frac{Hours}{Day} = 14560 \text{ Hours}$$

$$n_3 = 403 \text{ Rpm}$$

$$L_{10} = 60 \times h \times n_3 = 352 \, \text{Millions of Cycles}$$

Critical Bearing Force

$$A_p = 1.0 \text{ No Shock Present}$$

$$m = 3 \text{ Ball Bearing}$$

$$P_r = 6704 \text{ N}$$

$$X = 1.00 \text{ No Axial Force}$$

$$P = P \times X = \mathbf{6704\ N}$$

Left Bearing C

$$C_l = PL_{10}^{\frac{1}{3}} = \mathbf{7799\ N}$$

Right Bearing C

$$C_r = PL_{10}^{\frac{1}{3}} = \mathbf{47350\ N}$$

Bearing Selection

$$\text{Choose SKF } \mathbf{6407} : C = \mathbf{55300\ N}$$

Same Bearing Used for Both Sides

Left Bearing FOS

$$n_s = \frac{55300}{7799} = \mathbf{7.09}$$

Right Bearing FOS

$$n_s = \frac{55300}{47350} = \mathbf{1.17}$$